

$$3x + 7 = 22, \quad x=?$$

DeepSeek R1 (1041 tokens)

[Thinking Process]

<think>

Okay, so I need to solve the equation $3x + 7 = 22$. Hmm, let's think. I remember from class that to solve for x , I need to isolate it on one side of the equation. Right? So first, maybe I should get rid of that 7 that's being added to the term with x . <omitted> Starting with $3x + 7 = 22$. If I subtract 7 from both sides, it would be $3x + 7 - 7 = 22 - 7$. Simplifying both sides, the $+7$ and -7 on the left side cancel each other out, leaving just $3x$. And 22 minus 7 is 15. So now I have $3x = 15$. <omitted> Let me do that. Dividing both sides by 3 gives me $(3x)/3 = 15/3$. On the left side, the 3s cancel out, leaving just x . And 15 divided by 3 is 5. So $x = 5$. <omitted> But just to make sure I didn't make any mistakes in my steps. Let's go through it again. Original equation: $3x + 7 = 22$. Subtract 7 from both sides: $3x = 15$. Divide by 3: $x = 5$. Substitute back: $3*5 + 7 = 15 + 7 = 22$. Yep, that's correct. <omitted> If I start with $3x + 7 = 22$ and decide to divide everything by 3 first. Then I would have $x + 7/3 = 22/3$. Then subtract $7/3$ from both sides: $x = 22/3 - 7/3 = 15/3 = 5$. So same answer. So either method works. <omitted> Another thing to consider: what if I messed up the order of operations? Like, if I tried to subtract $3x$ first. Wait, but in the equation it's $3x + 7$, so the 3 is only multiplied by x , not the 7. So order of operations is correct here. <omitted> Wait, just to be thorough, could there be any other solutions? Let's see. Suppose someone thought that maybe x could be a different number. Let's test $x=4$. Then $3*4 + 7 = 12 + 7 = 19$, which is not 22. $x=6$ would be $18 + 7 = 25$, also not 22. So it's definitely 5. <omitted> The steps make sense, checked the solution, considered alternative methods, verified that the answer is unique. Yeah, I'm confident that $x=5$ is the right answer. </think>

[Final Response]

To solve the equation $3x + 7 = 22$:

Step 1: Subtract 7 from both sides

$$3x + 7 - 7 = 22 - 7 \implies 3x = 15$$

Step 2: Divide both sides by 3

$$3x/3 = 15/3 \implies x = 5$$

Verification:

Substitute $x = 5$ back into the original equation:

$$3(5) + 7 = 15 + 7 = 22 \quad \checkmark$$

Final Answer: 5

DeepSeek V3 (58 tokens)

[Final Response]

To solve the equation $3x + 7 = 22$ for x , follow these steps:

1. Subtract 7 from both sides:

$$3x + 7 - 7 = 22 - 7$$

Simplifying this, we get:

$$3x = 15$$

2. Divide both sides by 3:

$$3x/3 = 15/3$$

Simplifying this, we get:

$$x = 5$$

Final Answer: 5